

Sunil Banwari, during his 28+ years of working in the electronics industry, has been associated with the likes of Intel Corporation, Anadigics, Natel EMS, Smiths Group, ON Semiconductor, and AT&S India

Advantest Corporation is a leading manufacturer of automatic test equipment for the semiconductor industry, and a manufacturer of measuring instruments used in the design, production, and maintenance of electronic systems including fibre-optic and wireless communication equipment and digital consumer products. It recorded over \$3 billion in annual revenue in fiscal 2020.

Product Life Cycle. It is also sometimes referred to as the Technology Life Cycle. Industry 4.0 can benefit semiconductor manufacturing at every stage of the product or technology life cycle as it gives us an opportunity to do things in a smarter way.

Smart Manufacturing, when used as a synonym to Industry 4.0, includes smart design capabilities too. Designing a product today requires going through different iterations of the same. These different iterations go on until you hit the right spot and can meet all the expectations from the design. Typically, it takes multiple iterations to get one design right.

Industry 4.0 reduces the number of iterations dramatically. AI and ML being the biggest contributors play a big role here. The best example of the same is the recent announcement made by Google.

When you do a layout of a chip it requires a lot of designing, and the layout engineers have to ensure that the circuit is placed in a way that optimises the efficiency of a chip. Thankfully, Google has introduced a tool which can do the same using AI instead of humans.

According to Google, the tool de-

creases the time to market and helps bring down the cost by decreasing the amount of time and number of iterations required in the process. This was just an announcement by



Google and, personally, I do not think it has been adopted widely in the semiconductor industry. However, we should not forget that Google is designing its own chips.

There is a tool that we at Advantest offer that can decrease the time of validation of silicon. This tool is also based on Industry 4.0 and basically allows one to do a proper validation with every iteration and hence improve performance with every new iteration. It can even suggest design options on its own.

Any other elements where Industry 4.0 can be beneficial to the semiconductor industry?

Consider the example of an OSAT or an ATMP that manufactures chips for others but does not have its own design, say someone like the Taiwan Semiconductor Manufacturing Company (TSMC). The opportunity that Industry 4.0 provides to them helps improve the yield and capacity. Typically, the margins of such companies are lower than those that design the chips.

Industry 4.0 can provide the reasons behind losses in terms of capacity at each step for OSATs and ATMPs. It can also reduce the idle time caused by different reasons (including equipment not working) by suggesting what to do based on data it has collected in the past.

With much of the process being manual, it takes weeks to improve yield because the data is read, the equipment is reset and tested, before the final steps are taken. Industry 4.0 can automate the process, saving valuable time as the data is available with you at every point of time and process, and it does not need to be analysed manually.

What about the yield part?

As mentioned earlier, the usual process of semiconductor manufacturing, which used to be about two-and-a-half months long in the past, made it difficult as we had to wait to find out whether the die being used was good or not. It involved a lot of guesswork with the usual question being, "What percentage of what was started would be good

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